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FROM ALGORITHM TO COMPUTER PROGRAM

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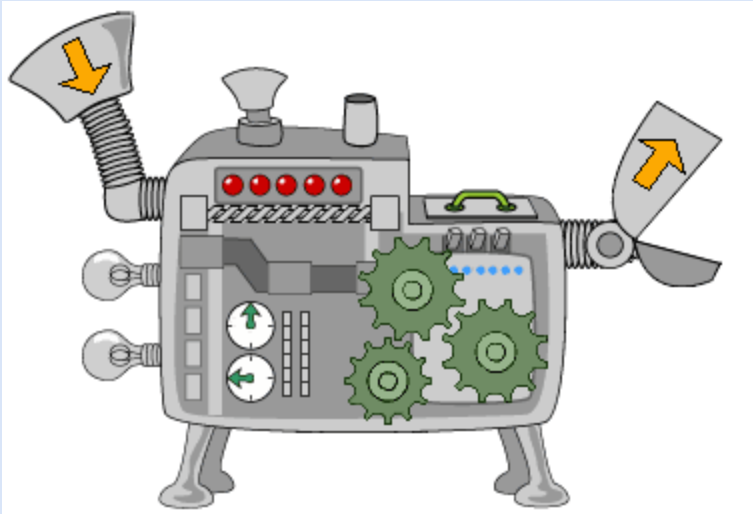
Contents

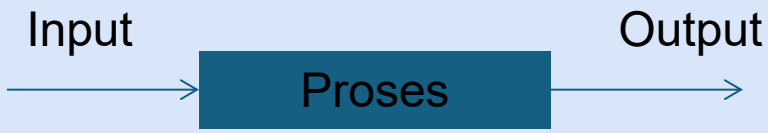
- Program examples
- Translating algorithm into C programming language
- C compiler



Input/Output Program

- A computer program generally receives input data and generates output.







Program Example

```
PROGRAM HelloWorld
{Program to write the text 'Hello, World!'. }
```

```
DECLARATION:
    { none }

ALGORITHM:
    Write('Hello, World!')
```

```
PROGRAM HelloWorld2
{Program to write the text 'Hello, World!'. }

DECLARATION:
    message : string

ALGORITHM:
    message ← 'Hello, World!'
    Write(message)
```



Program Example (cont.)

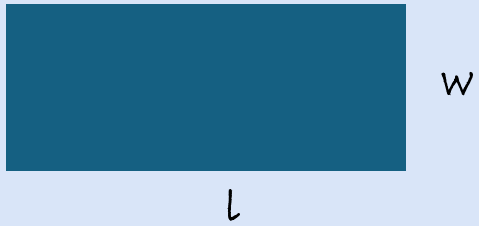
```
PROGRAM Greetings
{ Programs that accept input <name>
and print greeting 'Hello <name>'. }

DECLARATION:
    name : string

ALGORITHM:
    Read(name)
    Write('Hello ', name)
```



Program Example (cont.)




```
PROGRAM RectangularArea
{ The program reads the length (l) and width (w) of a rectangle, calculates
the rectangular area, then prints the result onto the screen.}
```

```
DECLARATION:
   l : real        { length of rectangular, in cm }
   w : real        { width of rectangular, in cm }
   Area : real     { area of rectangular, in cm² }
```

```
ALGORITHM:
   Read(l,w)       { input rectangular's length and width }
   Area ← l * w    { calculate rectangular area }
   Write(Area)     { print rectangular's area onto the screen }
```



Program Example (cont.)

```
PROGRAM CircleArea

{Given a circle with radius r, the program calculate the
area A of the circle. }

DECLARATION:
      Const Pi = 3.14     { Pi }
      r     : real        { radius of the circle }
      A     : real        { Area of the circle, Pi*r^2 }

ALGORITHM:

Read (r)                  { r > 0 }
A  ←  Pi * r * r          { calculate the circle area }
Write (A)
```



Task

1. Write a simple input-output program in the algorithm notation to calculate the area of the other geometry (triangle, trapezoidal, etc.). The input data is read from the input device and the area is displayed as an output.
2. Create a simple program in the algorithm notation that reads the temperature in Celsius, then converts it to Fahrenheit units, and displays the conversion to the screen. The formula from Celsius (C) to Fahrenheit (F) is $F = 9/5 \times C + 32$.
3. In a store, buyers get a discount of 5% price for each item purchased. Write an algorithm that reads the price of an item (in Rp.), then calculates the price of the item after it is discounted.
4. In conventional banks, the money saved in the amount of x gets interest at p% every year. Write the algorithm that reads x and p, then calculates the amount of money after being kept for one year.



Algorithm Pseudo-code Template

```
PROGRAM ProgramName
{ An explanation of the algorithm, which contains a brief
description of the problem that will be resolved, including
the input and output.}

DECLARATION:
{ This section contains declarations of all type names,
constants names, variables names, procedures names, and
functions names declared here.}

ALGORITHM:
{ This section is the core of the algorithm, contains
problem solving steps.}
```



C Program Template

```c
/* PROGRAM ProgramName */
/* An explanation of the program, which contains a brief description
of the problem that will be resolved, including the input and output.*/

/* DECLARATION: */
/* all names that are used globally declared here */

/* procedure and function prototypes declared here */

#include <stdio.h>

void main()
{
    /* DECLARATION: */
    /* all names that are used locally declared here */

    /* ALGORITHM: */
    /* all program's instructions are written here */
}
```



Notes on C language

1. The C language is case sensitive.
2. Comments are written between a pair of "/ *" and "* /".
3. `<stdio.h>` is the standard library for screen input / output commands.
4. Declarations made before the `main()` keyword are considered global declarations. Recognized throughout the program (not just in `main()`).



Writing a C Program

1. We can use a plain text editor program such as Notepad, Notepad++, or Sublime.
2. Or We could use an IDE such as Dev-C++.

3. A C program file is saved with the "**.c**" extension

4. Example : HelloWorld.c



Translation table

Group	Pseudo-code	C
Primitive type	boolean	Not available. We could use int, where true = 1, false = 0
	integer	- `int` - `short int` - `long int` - `unsigned int`
	real	- `float` - `double` - `long double`
	char	`char`
Composite type	string	Not available. We could make an array of char.

	record
	<field1 : type,
	field2 : type,
	…
	fieldN : type
	>
		`struct`
		`{ type field1;`
		`type field2;`
		`…`
		`type fieldN;`
		`}`



Translation table

Group	Pseudo-code	C
Operator: Arithmetics	+	+
	-	-
	*	*
	/	/
	div	/
	mod	%
Comparison	<	<
	≤	<=
	>	>
	≥	>=
	=	==
	≠	!=
Logical	not	!
	and	&&
	or	||
	xor	Not available



Translation table

Group	Pseudo-code	C
Comments	{ comments }	/* comments */
Etc	const	- const - #define
	type	typedef
	true	1
	false	0
Assignment	←	=
Reading	read	scanf
Writing	write	printf



Translating Ex. (HelloWorld)

Algorithm :

```
PROGRAM HelloWorld
{Program to write the text 'Hello, World!'. }

DECLARATION:
{ none }

ALGORITHM:
Write('Hello, World!')
```

C Program :

```c
/* PROGRAM HelloWorld */
/* Program to write the text 'Hello, World!' */

#include <stdio.h>

main()
{
    /* DECLARATION: */
    /* none */

    /* ALGORITHM: */
    printf("Hello, World! \n");
}
```



Translating Ex. (CircleArea)

Algorithm :

```
PROGRAM CircleArea
{Given a circle with radius r, the program calculate
the area A of the circle. }

DECLARATION:
    Const Pi = 3.14      { Pi }
    r    : real          { radius of the circle }
    A    : real          { Area of the circle }

ALGORITHM:
    Read (r)             { r > 0 }
    A  <- Pi * r * r     { calculate the circle area }
    Write (A)
```

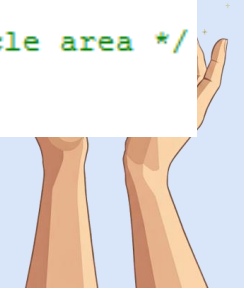
C Program :

```c
/* PROGRAM CircleArea */
/* Given a circle with radius r, the program calculate
the area A of the circle. */

#include <stdio.h>

main()
{
    /* DECLARATION: */
    #define Pi 3.14      /* Pi constant */
    float r;             /* radius of the circle */
    float A;             /* Area of the circle */

    /* ALGORITHM: */
    scanf("%f", &r);     /* r > 0 */
    A  = Pi * r * r;     /* calculate the circle area */
    printf("%f", A);
}
```



C Compiler

1. We use GCC to compile a C program.

2. To compile HelloWorld.c program :

Command Prompt

```
C:\PRG1>gcc HelloWorld.c -o HelloWorld.exe

C:\PRG1>HelloWorld
Hello, World!

C:\PRG1>
```



A Communicative Program

1. The user should know what to input and what is the output

```c
/* PROGRAM CircleArea */
/* Given a circle with radius r, the program calculate
the area A of the circle. */

#include <stdio.h>

main()
{
    /* DECLARATION: */
    #define Pi 3.14     /* Pi constant */
    float r;            /* radius of the circle  in cm*/
    float A;            /* Area of the circle in cm^2*/

    /* ALGORITHM: */
    printf("radius = ?"); scanf("%f", &r);  /* r > 0 */
    A  = Pi * r * r;    /* calculate the circle area */
    printf("Circle Area is %f cm^2", A);
}
```



A Communicative Program (cont.)

1. The user should know what to input and what is the output

```
C:\PRG1>gcc CircleArea2.c -o CircleArea2.exe

C:\PRG1>CircleArea2
radius = ?4
Circle Area is 50.240002 cm^2
C:\PRG1>
```



Task

Translate these algorithms to C program :

1. Write a simple input-output program in the algorithm notation to calculate the area of the other geometry (triangle, trapezoidal, etc.). The input data is read from the input device and the area is displayed as an output.
2. Create a simple program in the algorithm notation that reads the temperature in Celsius, then converts it to Fahrenheit units, and displays the conversion to the screen. The formula from Celsius (C) to Fahrenheit (F) is $F = 9/5 \times C + 32$.
3. In a store, buyers get a discount of 5% price for each item purchased. Write an algorithm that reads the price of an item (in Rp.), then calculates the price of the item after it is discounted.
4. In conventional banks, the money saved in the amount of x gets interest at p% every year. Write the algorithm that reads x and p, then calculates the amount of money after being kept for one year.



Useful links on C language

1. https://en.wikipedia.org/wiki/C_(programming_language)
2. https://www.tutorialspoint.com/cprogramming/
3. http://www.c-faq.com/



Bibliography

- Munir, Lidya, “Algoritma & Pemrograman”, 6th Ed, 2016 Penerbit Informatika.
- Kernighan, Ritchie, “The C Programming Language”, 2nd Ed, Prentice Hall.



Thank You!



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